

Iris Flower Classification using ML

Project Overview

This project was developed as part of the CodeAlpha Data Science Internship. The primary objective is to classify Iris flowers into different species using Machine Learning techniques based on their physical measurements.

The project focuses on the following flower species:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Objective

The main goals of this project are:

- Understand the fundamentals of classification in Machine Learning
- Explore and analyze the Iris flower dataset
- Build and train a Machine Learning model
- Evaluate the model's performance using standard metrics
- Predict flower species based on flower measurements

Dataset Information

The Iris dataset contains the following features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Variable:

- Species

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

- Scikit-learn
- Jupyter Notebook

Machine Learning Algorithm

K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a supervised Machine Learning algorithm widely used for classification tasks. It classifies a data point by analyzing the characteristics of its nearest neighboring data points.

Project Workflow

1. Import Libraries
2. Load Dataset
3. Data Understanding
4. Data Cleaning
5. Exploratory Data Analysis (EDA)
6. Data Visualization
7. Feature Selection
8. Train-Test Split
9. Model Building
10. Model Training
11. Prediction
12. Model Evaluation
13. Custom Prediction

Data Visualization

The project includes the following visualizations:

- Count Plot
- Pair Plot
- Correlation Heatmap
- Confusion Matrix Heatmap

These visualizations help in understanding feature relationships and the distribution of flower species.

Model Performance

The KNN model achieved high accuracy on the test dataset.

The following evaluation metrics were used:

- Accuracy Score
- Confusion Matrix
- Classification Report

Conclusion

This project successfully demonstrates the implementation of a Machine Learning classification model using the Iris dataset.

The K-Nearest Neighbors (KNN) algorithm performed effectively and achieved high accuracy in predicting flower species.

Through this project, I gained practical experience in:

- Data Preprocessing
- Exploratory Data Analysis
- Data Visualization
- Classification Algorithms
- Model Training
- Model Evaluation
- Prediction using Machine Learning

This project enhanced my practical understanding of Data Science and Machine Learning concepts while improving my analytical and problem-solving skills.